Supplementary Material

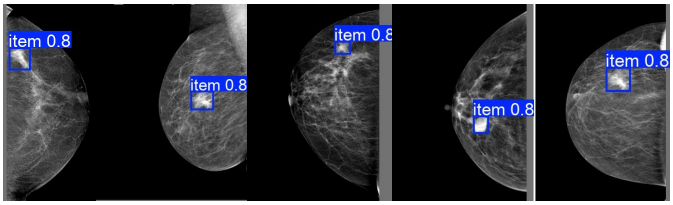


Fig. S1. Representative YOLOv11 detection results on the private in-house mammography dataset. Bounding boxes indicate expert-verified suspicious breast mass regions, with YOLOv11 achieving approximately 97.6% lesion localization success for reliable ROI generation before downstream segmentation, feature extraction, and molecular subtype classification.

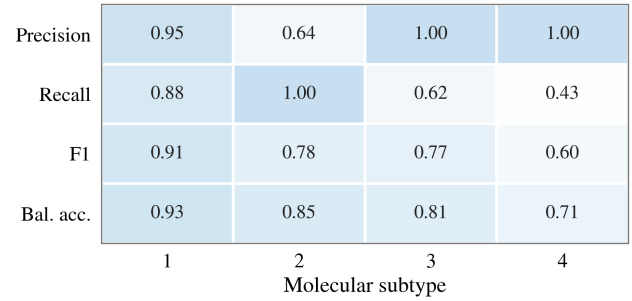


Fig. S2. Class-wise performance heatmap for the CatBoost molecular subtype classifier on Dataset B. The figure reports precision, recall, F1 score, and balanced accuracy for each subtype, providing a more detailed evaluation under class imbalance than aggregate ROC AUC alone. Columns 1–4 correspond to Luminal A, Luminal B, HER2-positive, and Triple-Negative disease, respectively.

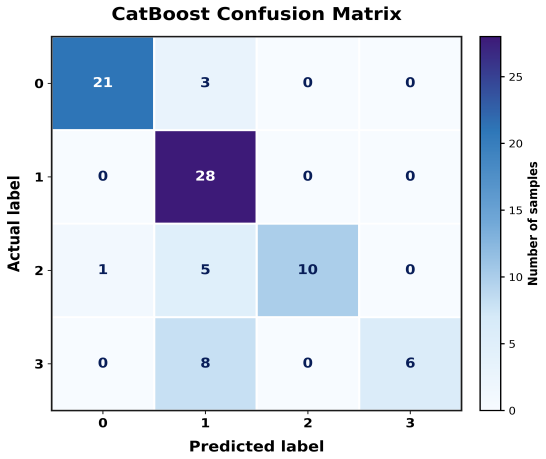


Fig. S3. Confusion matrix for the CatBoost molecular subtype classifier on Dataset B. Labels correspond to 0: Luminal A, 1: Luminal B, 2: HER2 positive, and 3: Triple Negative. The matrix provides class level insight into correct and incorrect predictions and highlights the greater variability observed for underrepresented aggressive subtypes.

References

[1] A. B. Panambur, P. Madhu, and A. Maier, "Classification of luminal subtypes in full mammogram images using transfer learning," *arXiv preprint arXiv:2301.09282*, 2023.

[2] M. A. Bakker, M. d. L. Ovalho, N. Matela, and A. M. Mota, "Decoding breast cancer: Using radiomics to non-invasively unveil molecular subtypes directly from mammographic images," *Journal of Imaging*, vol. 10, no. 9, p. 218, 2024.

[3] W. Ma, Y. Zhao, Y. Ji, X. Guo, X. Jian, P. Liu, and S. Wu, "Breast cancer molecular subtype prediction by mammographic radiomic features,"

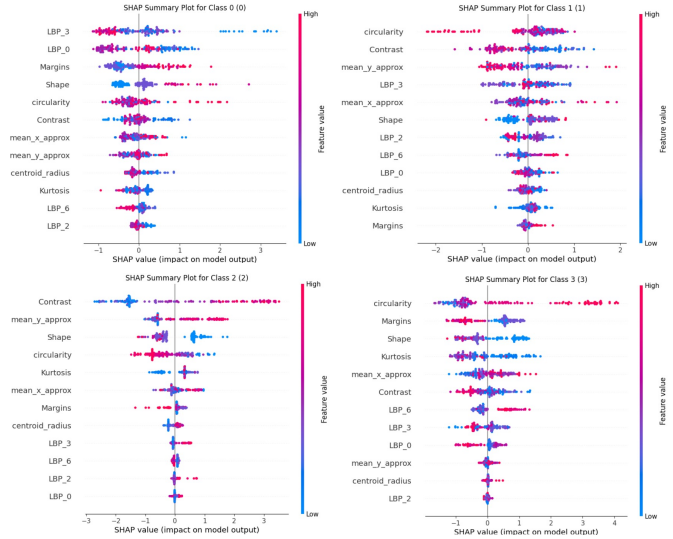


Fig. S4. Class specific SHAP summary plots for molecular subtype classification. For each class, features are ranked by mean absolute SHAP value, and their signed impact on the predicted probability is shown. Morphological descriptors, including shape, margin, and circularity, dominate across classes and reflect boundary dependent differences between subtypes. Contrast consistently ranks highly and is especially relevant for HER2 positive and Triple Negative predictions, while texture features such as LBP and kurtosis capture subtype specific heterogeneity.

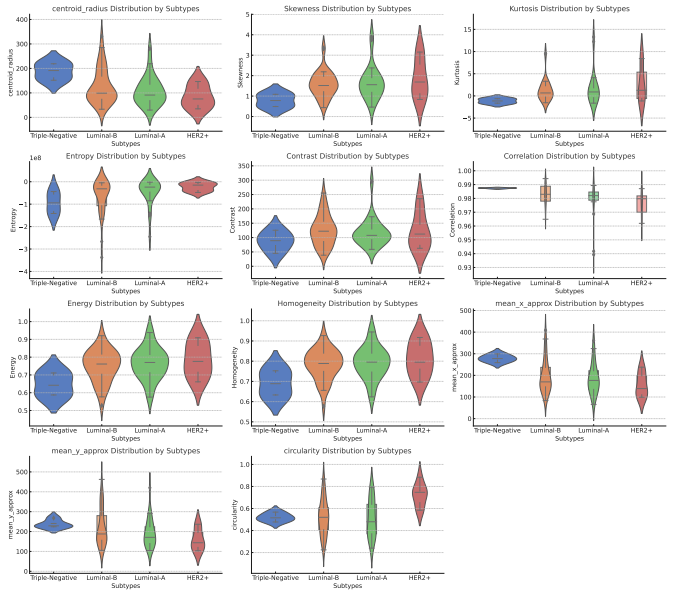


Fig. S5. Distribution of key morphological and texture features across molecular subtypes in the private cohorts. Box plots summarize median, interquartile range, whiskers, and outliers. The distributions provide descriptive support for the SHAP interpretation by showing subtype level variation in boundary geometry and intratumoral texture heterogeneity.

TABLE S1
CONTEXTUAL COMPARISON WITH REPRESENTATIVE PRIOR STUDIES ON BREAST CANCER MOLECULAR SUBTYPE PREDICTION. THIS COMPARISON IS LITERATURE LEVEL RATHER THAN A DIRECT HEAD TO HEAD BENCHMARK BECAUSE DATASETS, IMAGING MODALITIES, SUBTYPE DEFINITIONS, SAMPLE SIZES, AND VALIDATION PROTOCOLS DIFFER ACROSS STUDIES.

Study	**Imaging source**	**Approach**	**Subtype task**	**Reported performance**
Panambur et al. [1]	Mammography	Deep learning with transfer learning	Luminal versus non Luminal binary classification	AUC 0.6688
Bakker et al. [2]	Breast imaging radiomics	Radiomics feature based machine learning	Luminal A versus Luminal B; HER2 versus Triple Negative	AUC 0.714 and 0.755
Ma et al. [3]	Breast imaging radiomics	Radiomics feature based machine learning	HER2 versus non HER2; Triple Negative versus non Triple Negative	AUC 0.784 and 0.752
Son et al. [4]	Synthetic mammography from digital breast tomosynthesis	Radiomics with elastic net	Luminal, HER2, and Triple Negative prediction	AUC 0.838 for Triple Negative, 0.556 for HER2, 0.645 for Luminal
Wu et al. [5]	Mammography and ultrasound	White box machine learning using BI RADS features	Molecular subtype prediction	Interpretable imaging features reported
Mota et al. [6]	Mammography	Deep learning with transfer learning	Multiclass and binary molecular subtype prediction	Multiclass AUC 0.6062; binary AUC 0.7902
Singh et al. [7]	Mammography	VGGNet based deep learning	Multiclass and binary molecular subtype prediction	Accuracy 0.6700; binary accuracy 0.7801
Proposed method	Mammography	Boundary guided segmentation followed by morphology and radiomics enriched machine learning	Multiclass prediction of Luminal A, Luminal B, HER2 positive, and Triple Negative	Dataset A: AUC 0.946, accuracy 0.843; Dataset B: AUC 0.937, accuracy 0.802

TABLE S2
OPTIMAL HYPERPARAMETERS FOR THE MACHINE LEARNING CLASSIFIERS.

Classifier	**Hyperparameter**	**Value**
Random Forest	Max depth	7
	Min samples leaf	1
	Min samples split	2
	Number of estimators	200
XGBoost	Learning rate	0.1
	Number of estimators	150
	Gamma	0.3
	Booster	gbtree
CatBoost	Iterations	200
	Learning rate	0.1
	Depth	7
	Verbose	0
MLP	Max iterations	250
	Activation	relu
	Solver	adam
	Batch size	64

TABLE S3
SUMMARY OF KEY WORKFLOW PARAMETERS REPORTED FOR REPRODUCIBILITY.

Workflow step	Reported setting
ROI definition	YOLOv11 candidate lesion localization followed by a fixed 5-pixel bounding-box expansion in each direction and ROI cropping
Segmentation input size	256×256 pixels
Intensity preprocessing	Histogram equalization and normalization to [0,1]
Data augmentation	Rotations at 0°, 90°, 180°, and 270°; CLAHE-based contrast variation
Segmentation loss	Cross-entropy + Dice; $\lambda_1 = 0.5$, $\lambda_2 = 0.5$; weights fixed before held-out test evaluation
Feature extraction region	Segmented tumor mask applied to the mammographic ROI
Morphological descriptors	Area, perimeter, circularity, centroid radius, contour approximation, convex hull-based descriptors
Texture descriptors	Contrast, entropy, homogeneity, energy, skewness, kurtosis, and LBP with $R = 1$, $P = 8$
Feature selection	`SelectKBest` with `mutual_info_classif`, applied within training folds; $K = 9$ for shape/margin and $K = 12$ for subtype classification
Class balancing	SMOTE fitted within the training fold only; MT resampled in MT space and MT+SM in the corresponding fused training representation; minority classes targeted to the majority-class count (ratio 1.0); no SMOTE on held-out data or Dataset B
Sequential phenotype inputs	Held-out samples receive predicted, not reference, shape and margin labels so that upstream errors are propagated into subtype evaluation
Model selection	5-fold cross-validation and GridSearchCV for hyperparameter tuning
Domain-shift handling	Common preprocessing and supervised segmentation fine-tuning; no dedicated adversarial/feature-alignment domain-adaptation objective; no formal ComBat harmonization
External evaluation	Dataset B evaluated without additional fine-tuning of the downstream subtype classifier

Academic radiology, vol. 26, no. 2, pp. 196–201, 2019.

[4] J. Son, S. E. Lee, E.-K. Kim, and S. Kim, "Prediction of breast cancer molecular subtypes using radiomics signatures of synthetic mammography from digital breast tomosynthesis," *Scientific reports*, vol. 10, no. 1, p. 21566, 2020.

[5] M. Wu, X. Zhong, Q. Peng, M. Xu, S. Huang, J. Yuan, J. Ma, and T. Tan, "Prediction of molecular subtypes of breast cancer using bi-rads features based on a "white box" machine learning approach in a multi-modal imaging setting," *European journal of radiology*, vol. 114, pp. 175–184, 2019.

[6] A. M. Mota, J. Mendes, and N. Matela, "Breast cancer molecular subtype

TABLE S4
COMPLETE CANDIDATE FEATURE SET USED FOR MOLECULAR SUBTYPE CLASSIFICATION.

Feature group	Specific features
First-order intensity	Mean Intensity, Variance, Max Intensity, Skewness, Kurtosis
Texture	Entropy, Contrast, Correlation, Energy, Homogeneity
Local texture	LBP_0–LBP_9
Morphological / contour	centroid_radius, perimeter, area, circularity, mean_x_approx, mean_y_approx, mean_x_hull, mean_y_hull, std_x_approx, std_y_approx, std_x_hull, std_y_hull
Predicted phenotypes	Shape, Margins

Note: The first 32 variables are image-derived candidate descriptors. Shape and Margins are predicted intermediate phenotypes included only in the augmented MT+SM setting. SelectKBest with mutual information was applied within each training fold, retaining the top 12 predictors for molecular subtype classification.

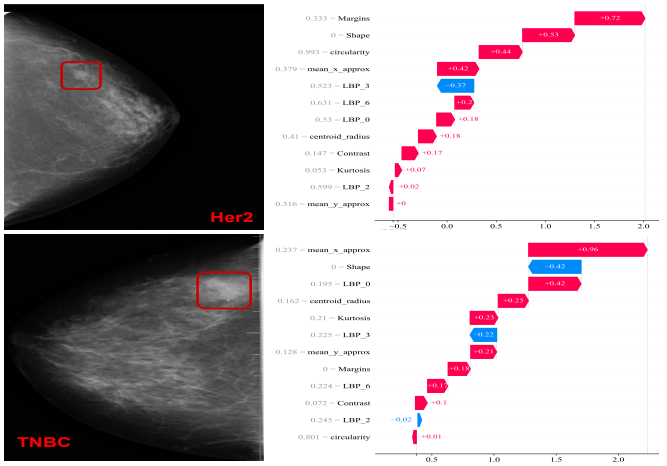


Fig. S6. Representative SHAP explanations for misclassified aggressive subtype cases. These examples illustrate failure behavior by showing how individual morphology and texture features shift the model output away from the true subtype. The analysis supports a cautious interpretation of SHAP as an associative explanation tool and highlights the need for clinical review in uncertain or atypical cases.

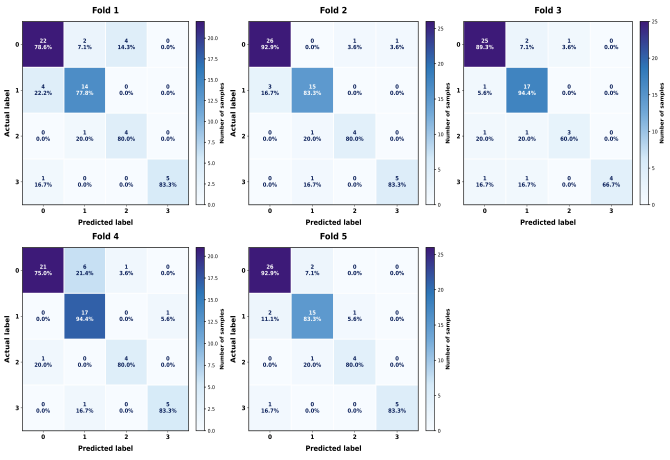


Fig. S7. Fold-wise confusion matrices for the CatBoost classification model, showing the distribution of predicted and actual class labels across each cross-validation fold.

prediction: A mammography-based ai approach," *Biomedicines*, vol. 12, no. 6, p. 1371, 2024.

[7] V. K. Singh, S. Romani, J. Torrents-Barrena, F. Akram, N. Pandey, M. M. K. Sarker, A. Saleh, M. Arenas, M. Arquez, and D. Puig, "Classification of breast cancer molecular subtypes from their micro-texture in mammograms using a vggnet-based convolutional neural network," in *Recent Advances in Artificial Intelligence Research and Development*. IOS Press, 2017, pp. 76–85.

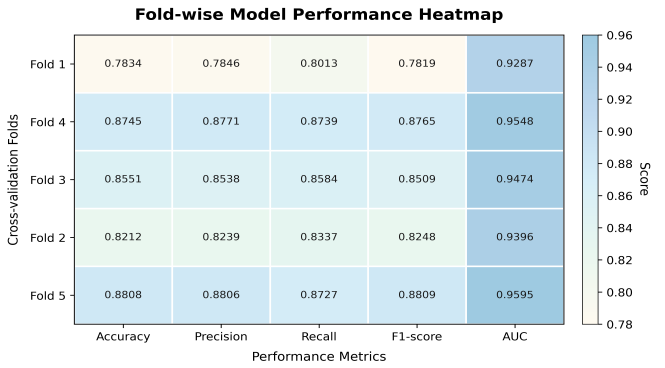


Fig. S8. Heatmap of fold-wise performance metrics, including accuracy, precision, recall, F1-score, and AUC, for the CatBoost classification model across cross-validation folds.

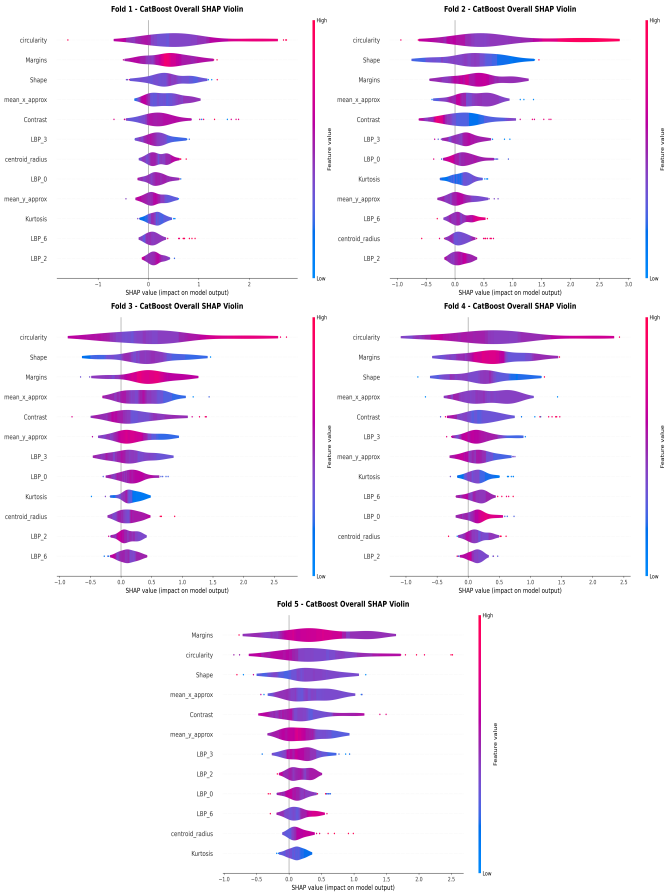


Fig. S9. Fold-wise SHAP analysis showing stable and reproducible feature contributions across cross-validation folds, confirming that the molecular subtype classifier relies on consistent boundary, morphological, and texture-based mammographic descriptors.